

Harsh Raj

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Professional Summary

Final-year Computer Science student with experience developing AI-powered applications, developer tools, and full-stack web platforms. Skilled in Python, JavaScript/TypeScript, React, FastAPI, PostgreSQL, and cloud deployment, with research experience in automated code refactoring and software quality evaluation.

Education

VIT-AP University, Amaravati

B.Tech in Computer Science and Engineering, SCOPE

CGPA: 8.56 / 10

Expected: September 2026

Vidya Bharati Chinmaya Vidyalaya, Jamshedpur

Senior Secondary School

Class 10: 94.4% Class 12: 83.2%

March 2022

Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, HTML/CSS.

Frameworks & Libraries: React, Next.js, FastAPI, Flask, TailwindCSS.

Databases: PostgreSQL, Supabase, Drizzle ORM.

Tools & Platforms: Git, GitHub, AWS, Vercel, Railway, Google Cloud, N8N.

AI/ML: Machine Learning, NLP, Google API, OpenAI API, Prompt Engineering.

Projects

- **Argos - Competitive Live Debate Platform** [Live Link]
Tech: Next.js, TypeScript, Supabase, PostgreSQL, Drizzle ORM, Gemini AI, TailwindCSS, Vercel
Built a full-stack debate platform supporting 2–5 round competitive matches with AI evaluation across 4 scoring dimensions and detection of 10 logical fallacies. Engineered 15+ API routes, 8 database tables, real-time multiplayer synchronization, Google OAuth authentication, and Elo-based ranked matchmaking. Integrated Gemini AI for automated scoring and feedback generation, while deploying a scalable cloud architecture with public debate feeds, challenge lobbies, daily AI-generated topics, and automated debate lifecycle management.
- **MergeIQ - AI-Powered Pull Request Analyzer** [GitHub] | [Live Link]
Tech: Python, FastAPI, Chrome Extension (MV3), Gemini 2.5 Flash, GitHub REST API, PyJWT, Railway
Built a Chrome extension and GitHub App that automatically analyzes every pull request using Gemini AI, delivering structured verdicts (Safe / Review Needed / Risky) with risk flags, summaries, and suggested review questions. Fetches live repo context (README, folder structure, dependencies, recent commits) for project-aware analysis. GitHub App mode posts analysis comments autonomously on PR open via webhook, requiring zero manual interaction from contributors.
- **HealthAssist - AI Symptom Checker and Health Companion** [Live Link]
Tech: Python, FastAPI, React, TailwindCSS, Google API, Generative AI, Render, PostgreSQL
Implemented a web-application that generates detailed analyses of the symptoms provided by the patient including blood-report context-awareness, with personalized advices and precautions. Integrated 4 key data-points with 5 levels of analyses/advices.

- **CodeSage - AI code-base refactoring system and questions generator** [Live Link]
Tech: Python, Flask, React, TailwindCSS, Google API, Google Cloud, OpenAI API, LLMs
 Implemented an AI code-base analysis and refactoring system that helps improve code quality, maintainability, and generates insights about your projects using Google's Gemini AI or other LLMs. Supports more than 15 programming languages. Conducted a detailed research report on the performance of 5 leading LLMs with different prompts and scoring metrics. Currently developing a cAST-based pipeline for cutting-edge refactoring capabilities.

Research

Performance Evaluation of LLMs and Prompt Strategies for Automated Python Code Refactoring:

Evaluated 4 frontier LLMs and 2 prompt-engineering strategies for automated Python code refactoring across 5 benchmark programs. Measured code quality and documentation improvements using BLEU, ROUGE-L, and Pylint metrics, achieving style-conformance scores of 9.2–9.7/10. Conducted comparative analysis of model and prompt performance to identify effective refactoring strategies.

Certifications

- **Data Analysis in R (DataCamp)** [Link]
- **MERN Full Stack (Ethnus)** [Link]
- **AWS Certified Cloud Practitioner (AWS)** [Link]
- **Oracle OCI Generative AI Professional (Oracle)** [Link]
- **Oracle Certified Foundations Associate (Oracle)** [Link]
- **JavaScript Essentials (Cisco)** [Link]
- **Networking Essentials (Cisco)** [Link]

Achievements

- **2nd Place, The Return Journey Hackathon** (GDSC, IIT-Kanpur)

Leadership & Roles

- **Manager, TEAM NEXT NEXUS** (Prompt Engineering Club) July 2024 – December 2024
 Led prompt-engineering study groups and reviewed generative-AI mini-project submissions, facilitating collaborative learning and technical discussions among club members.
- **Associate Manager, Madhya Bharat Association** (Cultural Club) July 2024–December 2024
 Oversaw event logistics and team coordination for cultural events with 50+ participants.

Additional Information

Languages: English (Proficient), Hindi (Native)

Interests: AI Research, Number Theory, Art/Animation, Content Creation